

Exercise Sheet 6 - Linear Discriminant Analysis

Deadline for submission: Wednesday, 12 November 2025 at 8:00.

Discussed in exercise session: Friday, 14 November 2025 at 10:15.

Exercise 6.1 - LDA on Synthetic Data

Before applying an algorithm to real data, it's usually best practice to first implement and evaluate it for some synthetically created data (often called “toy data”). In this exercise, you will implement LDA, and check whether your method works by evaluating it on a synthetic data set.

More specifically, you will estimate the projection vector w for LDA with the multiple different approaches covered in the lecture.

Complete the tasks in the Jupyter Notebook `LDA_toy_empty.ipynb`

Secondary Structure Prediction of Proteins

The lecture introduced the problem of secondary structure prediction for proteins as an application for the Fisher LDA. Specifically, we are trying to predict the existence and position of α -helices in new proteins $y_i \in \{-1, 1\}$, given data in form of sequences with the corresponding 'Helix' annotations, i.e. we're in a classification setting.

For the 'features' (i.e. input x_i) for each y_i we will look at a sliding window of 5 amino acids with y_i corresponding to the class of the amino acid in the center. To actually encode these 5 amino acids, we will simply count the occurrence of each amino acid and form a vector with them, i.e. $x_i \in \mathbb{R}^{21}$ for 21 distinct amino acids.

In this exercise, we will apply this approach to a real data set, obtained from Kaggle¹. The file of interest is `2018-06-06-pdb-intersect-pisces.csv`. After downloading the file you can proceed with the provided Jupyter Notebook, where the required preprocessing of the data is handled.

Complete the tasks in the Jupyter Notebook `protein_structure_empty.ipynb`

Exercise 6.2 - Building up the required Matrices

Similar to the lecture, we would now be dealing with a matrix $X \in \mathbb{R}^{N \times 21}$, where N will become very large. To avoid potential memory issues, we will build the required Matrices/Vectors $X_c^T X_c$ and $X^T \mathbf{y}$ for solving the FLDA iteratively (compare Chapter 02 Slides 53+).

Complete the corresponding parts in the provided Notebook: ²

1. Build up the Matrices/Vectors $X^T X$, $X^T \mathbf{y}$ and $n\mathbf{m}$
2. Use them to calculate $X_c^T X_c$

Exercise 6.3 - Fisher LDA with SVD

Complete the corresponding parts in the provided Notebook:

1. Given the required matrices, estimate $\hat{w}$ for Fisher LDA using the SVD.
2. Given this vector $\hat{w}$, predict the structure of your 'test data set' by projecting the x_i onto the 1-dim subspace. On this subspace we can then predict the class based on a fixed threshold ($w^t x_i \geq t$). Try different thresholds and compare the predictions with the true $\mathbf{y}_{test}$.

¹<https://www.kaggle.com/alfrandom/protein-secondary-structure>

²If you have problems here, you can skip this part, load our provided solutions for the two matrices, and proceed with part 2.

3. What can you in general say about the choice of thresholds? Would your choice be different in a high risk setting such as the prediction of a disease, compared to a movie recommendation?

Elaborate in context of type-1 and type-2 errors.³⁴

³https://en.wikipedia.org/wiki/Type_I_and_type_II_errors

⁴(type 1 and type 2 errors, as well as the choice of thresholds are not relevant for exam)